DATASHEET

HALCON Calibration Plate, Aluminium Composite

MVTec HALCON calibration plate. A central reference circle is surrounded by smaller dot marks arranged in a regular grid. The asymmetric mark sizes allow `find_caltab` to determine board orientation without additional fiducials. Designed for use with the `find_caltab` and `calibrate_cameras` operators in HALCON 12 and later.

1 Ordering

Part number	Size (W × H)	Mark grid	Central mark ⌀	Grid mark ⌀	Total marks
CT-HAL-P-200X150-STD	200 × 150 mm	7 × 5	8 mm	3 mm	36
CT-HAL-P-300X200-STD	300 × 200 mm	7 × 7	10 mm	4 mm	50
CT-HAL-P-400X300-STD	400 × 300 mm	9 × 7	12 mm	5 mm	64
CT-HAL-P-600X400-STD	600 × 400 mm	11 × 9	14 mm	6 mm	100
CT-HAL-P-800X600-STD	800 × 600 mm	13 × 11	16 mm	7 mm	144

2 Key specifications

ACCURACY

±0.3 mm + 0.2 mm/m (any mark centre to nominal grid, at 20 °C)

SUBSTRATE

Aluminium / LDPE / aluminium composite, 6 mm

FLATNESS

≤ 0.5 mm per metre, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

1 Specifications · geometry

FEATURE	SPECIFICATION
Feature position tolerance	±0.3 mm + 0.2 mm/m (any mark centre to nominal grid, at 20 °C)
Mark pitch, mean	±0.05 mm
Circularity	≤ 0.05 mm radial deviation
Edge radius	R6 mm

2 Specifications · physical

FEATURE	SPECIFICATION
Substrate	Aluminium / LDPE / aluminium composite, 6 mm
Weight	6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg)
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per metre, board supported on three points
Mounting	4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size

3 Specifications · environment

FEATURE	SPECIFICATION
Operating temperature	−50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Storage	−20 to +60 °C, dry
Outdoor lifetime	up to 3 years (UV-stable print)
Cleaning	water or ≤ 70 % isopropyl alcohol, soft cloth

4 Deliverables

INCLUDED		SERVICE	
Measurement	Every board is measured before it ships (dimensional, flatness, visual)	Lead time	Made to order · ships in 2–3 weeks
Files	`.halcon.descr`, `.opencv.json`, `.dxf`	Options	Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE
Documents	this datasheet, CT-UM-GEN user manual		

5 Compliance & customs

TOPIC	STATEMENT
RoHS	RoHS-compliant materials.
HS code	`9031 80` Not a toy. not for food contact.

6 Revision history

REV	DATE	CHANGE
A	`2026-10`	First issue, planning values

7 Disclaimer

Values are nominal unless stated as measured.